



MediCare

// Console-Based Hospital Management System

Java 17 · MongoDB Atlas · Maven · Layered Architecture

Department of Computer Science & Engineering

SRM University

Prabhat Arora | Reg. No.: 06 | AI/ML A

Subject: Advanced Programming Practice

PROBLEM STATEMENT

⚠ Hospitals still rely on manual, paper-based systems for appointment booking and prescription management — leading to inefficiencies, scheduling conflicts, and poor patient experience.

24+

days avg wait
time

30%

scheduling
conflicts

15%

prescriptions
lost

60%

admin time
wasted

WHO IS AFFECTED

Doctors

Cannot efficiently manage time slots & patient flow

Patients

Face long wait times, no real-time availability info

Admin Staff

Struggle with record-keeping & scheduling coordination

KEY PAIN POINTS

- ▶ Scheduling conflicts with overlapping appointments
- ▶ Lost or misplaced paper prescriptions
- ▶ No real-time doctor availability for patients
- ▶ Time-consuming manual record management
- ▶ Difficulty tracking patient history

ABSTRACT - OUR SOLUTION

MediCare is a console-based, menu-driven application that digitizes hospital appointment booking and prescription management. Built with Java 17 and MongoDB Atlas, it enables doctors to publish time slots, patients to book appointments in real-time, and generates digital prescriptions — all within a strict modular, layered architecture.

// CORE MODULES

Appointment Booking

Browse doctors, check slots,
book & cancel appointments

Doctor Management

Publish availability, manage
schedules & patient flow

Prescription System

Digital prescriptions linked
to completed appointments

User Auth & Roles

Secure login, role-based access
for Doctors & Patients

// HOW IT WORKS

01

Register

02

Login

03Browse
Doctors**04**Book
Slot**05**

Consult

06

Prescribe

ARCHITECTURE & TECH STACK

// LAYERED ARCHITECTURE

UI Layer

Main.java - Menu-driven console

Service Layer

Business logic & validation

Repository Layer

Raw MongoDB CRUD operations

Connection Layer

Singleton MongoDB Atlas link

// TECH STACK

Java 17

LTS / Strong type safety / Platform independent

MongoDB Atlas

Cloud NoSQL / Scalable / Built-in security

Apache Maven

Build automation / Dependency management

Git & GitHub

Version control / Branching strategy

// DATA MODELS

User.java

Abstract base
ID, name, email, password

Doctor.java

Extends User
Appointment IDs list

Patient.java

Extends User
Appointment IDs list

Appointment.java

ID, doctorId, patientId
date, slot, status

Prescription.java

ID, diagnosis, medicines
timestamp

WHY IT'S BETTER

METRIC	BEFORE (Manual)	AFTER (MediCare)
Appointment Booking	Phone calls, walk-ins, 24+ day wait	Real-time digital booking, instant confirmation
Scheduling	30% conflicts, double-bookings	Zero conflicts, slot validation before booking
Prescriptions	15% lost, paper-based	100% digital, linked to appointments
Data Storage	Paper files, cabinets	MongoDB Atlas cloud, 99.9% uptime
Access	Hospital hours only	24/7 cloud access from anywhere
Record Tracking	Manual search, lost files	Instant lookup, full patient history

// IMPACT

30%

fewer scheduling
conflicts

100%

digital prescription
coverage

10x

faster booking
process

24/7

cloud database
availability



session – medipresenter

THANK YOU

Questions & Answers

```
>>> medipresenter --status
```

```
[OK] Presentation delivered successfully
```

```
[OK] All modules demonstrated
```

```
>>> _
```

MediCare — Hospital Management System

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